

Pradeep Gandu

Data Engineer | Cloud Data Engineer | AI/ML | Automotive Software

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PROFESSIONAL PROFILE

Automotive Software Engineering master's student with prior professional experience in software, structured-data, quality-oriented and technical project environments. Hands-on experience with data engineering workflows using Python, SQL, PySpark, Spark SQL, GCP and distributed data processing, with additional AWS cloud exposure. Current work includes CARLA-based autonomous-driving dataset generation, cloud data pipelines and local AI applications.

CORE TECHNICAL SKILLS

Data Engineering Python, SQL, PySpark, Spark SQL, ETL/ELT, Data Pipelines, Data Quality, Schema Validation, JSON, Parquet, HDFS	Cloud GCP, BigQuery, Pub/Sub, Cloud Storage, Dataproc, Compute Engine, IAM, gcloud, gsutil; AWS: EC2, S3, IAM, RDS, VPC, CloudWatch
AI / ML YOLOv8, Computer Vision, Pandas, NumPy, Ollama, Qwen3, Whisper, Model Evaluation	Automotive / Software CARLA, ADAS scenarios, trajectories, dataset annotation, FastAPI, Streamlit, Pydantic, Pytest, Git/GitHub, Linux, SSH

PROFESSIONAL EXPERIENCE

Working Student - AI-Assisted Software Solutions

Jan 2025 - Jul 2025

DiPP GmbH | Germany

- Worked with AI-assisted software solutions and technical-information processing workflows.
- Analysed technical workflows, outputs and error patterns; supported structured documentation.

Translator - India Formatting Team

Dec 2021 - Jul 2022

Ascent Staffing Solutions Pvt. Ltd. (client: Clarivate Analytics) | India

- Worked with structured technical information in quality-focused processing and formatting workflows.

Associate Language Specialist, Content Engineering

Oct 2020 - Dec 2021

GlobalLogic Technologies Private Ltd. | India

- Worked with structured datasets, data preparation, validation, data-quality checks and systematic error analysis.
- Supported analysis-oriented workflows and technical documentation.

Senior Process Executive

Sep 2017 - May 2019

Cognizant Technology Solutions India Private Ltd. | Hyderabad, India

- Worked with structured business, process and system data, process tracking and quality controls.

Practitioner - CRM Operations

Mar 2016 - May 2017

Concentrix Daksh Services India Private Limited | India

- Worked in structured digital and process-oriented operational workflows with quality and issue-analysis tasks.

Associate Software Engineer

Jun 2013 - Jun 2015

Magus IT Solutions Pvt. Ltd. | Hyderabad, India

- Gained professional experience in a software-engineering environment and supported technical project activities.

SELECTED TECHNICAL PROJECTS

Automotive Telemetry Streaming Platform - GCP

GCP | Pub/Sub | BigQuery | Cloud Storage | Python | SQL | IAM

- Designed a cloud-based architecture for ingesting and storing vehicle telemetry.
- Created partitioned and clustered BigQuery tables, Pub/Sub resources and a separate path for invalid events.

Dataproc / PySpark Retail Data Pipeline

PySpark | Spark SQL | Dataproc | GCS | HDFS | JSON | Parquet

- Built an ETL workflow over customer, order and salesman datasets using Spark DataFrames and SQL.
- Converted JSON to Parquet, combined regional data with UNION, joined business entities and generated an aggregated Retail Summary.

SELECTED TECHNICAL PROJECTS - CONTINUED

Scenario-Centric Autonomous Driving Dataset Pipeline - Master Thesis

[Python](#) | [CARLA 0.9.15.2](#) | [Pandas](#) | [YAML](#) | [JSON](#) | [Data Validation](#)

- Developing a CARLA-based pipeline for highway lane-change and merge scenarios.
- Generates structured scene/frame annotations, trajectories, lane information and quality checks for environment-prediction datasets.

YOLOv8 Bicycle Detection

[Python](#) | [YOLOv8](#) | [Computer Vision](#) | [NVIDIA GPU](#)

- Prepared and validated approximately 1,000 images and trained an object-detection model.
- Reached approximately 0.51 mAP50 and analysed inference results and error cases.

Local AI Applications

[Python](#) | [FastAPI](#) | [Streamlit](#) | [Ollama](#) | [Qwen3](#) | [Whisper](#) | [Pydantic](#) | [JSON Schema](#)

- Built an AI interview-practice workflow with speech input, local LLM processing and text-to-speech feedback.
- Developed a CV-to-job matching application using structured LLM outputs, validation and rule-based fallbacks.

EDUCATION

Master - Automotive Software Engineering
Technische Universität Chemnitz, Germany

In progress
90 / 120 ECTS

M.Tech - Embedded Systems
MLR Institute of Technology, India

2017 - 2019
First Class, CGPA 7.33

B.Tech - Electrical & Electronics Engineering
Jawaharlal Nehru Technological University Hyderabad, India

2009 - 2013
First Class, 62.67%

CERTIFICATIONS & LANGUAGES

Programs/Certifications: Google Cloud Data Engineering Program; Python Essentials 1 & 2 - Cisco/OpenEDG; SQL Basic - HackerRank; Goethe-Zertifikat B1.

Languages: German - Goethe B1 certified; English - IELTS B2; Telugu - first language.

Current thesis: Generation and Annotation of Scenario-Centric Datasets for Environment Prediction in Autonomous Driving.